

Rajarshi Chakraborty

Portfolio | GitHub | LinkedIn | LeetCode

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EDUCATION

- Techno Main Salt Lake** Kolkata, India
Bachelor of Technology - Computer Science and Engineering; GPA: 8.01 *July 2023 – Present*
Courses: Operating Systems, Data Structures and Algorithms, Database Management Systems, Computer Networks, Object-Oriented Programming, Software Engineering, Analysis of Algorithms

SKILLS

- Languages:** C/C++, Java, JavaScript, TypeScript, SQL, Bash
- Frameworks:** React, Next.js, TailwindCSS, ShadCN, Node.js, Express.js, LangChain, BetterAuth
- Tools:** Git, Docker, PostgreSQL, MySQL, MongoDB, Prisma, Mongoose, Drizzle, Postman, CI/CD Pipelines
- Platforms:** Linux, Windows, Supabase, Vercel

EXPERIENCE

- Techno Main Salt Lake** Kolkata
Undergraduate Researcher *2026 – Present*
 - Time-Series Modeling & Forecasting:** Developed **LSTM, ARIMA, and RNN-based models** for weather prediction using **time-series analysis** on real-world datasets, leveraging **statistical modeling** and **deep learning** techniques.
 - Production-Ready ML Systems:** Built **scalable, deployment-ready ML pipelines** using **Python, NumPy, Pandas, Scikit-learn, PyTorch, Node.js**, and time-series frameworks for end-to-end implementation.

PROJECTS

- ExeOS-AI** - AI-Powered Executive Operating System [\[Demo\]](#) | [\[GitHub\]](#) *Next.js, TypeScript, React, PostgreSQL, Drizzle ORM, Vercel AI SDK*
 - Engineered a production-grade **AI executive workspace** unifying email, calendar, tasks, and notes using **Next.js 16 App Router, React 19 Server Components**, and **TypeScript**.
 - Built an **AI-powered productivity engine** for email analysis, conversation summarization, and contextual recommendations by orchestrating multiple frontier **LLMs** (OpenAI, Anthropic Claude, Google Gemini) through **OpenRouter** and the **Vercel AI SDK**.
 - Developed an **event-driven backend** with **Inngest** for asynchronous AI jobs, scheduled automations, and long-running background workflows.
 - Implemented secure **multi-user authentication** and organization management with **Clerk**, and optimized **PostgreSQL** queries via **Drizzle ORM** for low-latency data retrieval.
 - Designed a responsive, accessible **SaaS dashboard** using **shadcn/ui**, **Tailwind CSS**, and **Zod** validation with a modular, type-safe architecture.
- Zenith CLI** - AI-Powered Developer Workspace & CLI [\[Demo\]](#) | [\[GitHub\]](#) *TypeScript, Next.js, PostgreSQL, Prisma, AI SDK*
 - Engineered a **decoupled AI developer workspace**: a **Bun**-powered terminal CLI paired with a **Next.js** glassmorphic web dashboard for full-stack local-to-cloud tooling.
 - Implemented secure **CLI-to-Web authentication** via the **OAuth 2.0 Device Authorization Flow (RFC 8628)** and **Better Auth**, enabling **GitHub OAuth** sign-in from the terminal.
 - Built an **autonomous agentic scaffolder** using the **Vercel AI SDK** and **Gemini API** with **Function Calling** pipelines for sandboxed code execution and **RAG**-based context injection.
 - Reduced **CLI startup latency** via **ESM** bundling with **tsup/esbuild** and lazy-loaded service architecture.
- SynapseUI** - React UI Component Library [\[Demo\]](#) | [\[GitHub\]](#) *Next.js, TypeScript, Tailwind CSS, Fumadocs*
 - Built a **production-ready, open-source UI component library** for **React** and **Next.js**, supporting **copy-paste** integration or install via **ShadCN CLI**.
 - Architected a **fully customizable, type-safe component system** using **TypeScript**, styled with **Tailwind CSS** and **ShadCN** primitives.
 - Delivered comprehensive **documentation** powered by **Fumadocs** with live component previews and CLI-based scaffolding.
- SocketPoll** - Real-Time Polling App [\[Demo\]](#) | [\[GitHub\]](#) *React, Express.js, Socket.IO, Node.js*
 - Designed a **low-latency, event-driven polling platform** using **WebSocket (Socket.IO)** for **bidirectional real-time communication** and instant vote propagation.
 - Built a **RESTful Express.js backend** with **room-based namespace isolation** and **in-memory state management** for concurrent sessions.
 - Developed a **reactive Vite + React frontend** with **optimistic UI updates** and animated result visualization.
- LinkedUp** - LinkedIn Post Generator [\[Demo\]](#) | [\[GitHub\]](#) *Python, LangChain, OpenAI API, Streamlit*
 - Engineered an **LLM-powered content generation pipeline** using **LangChain** for **prompt chaining**, **retrieval-augmented generation (RAG)**, and output parsing.
 - Integrated **few-shot prompting** and **memory buffers** for multi-turn coherence with real-time **token streaming**.

HONORS & CODING PLATFORMS

- Solved **650+** problems on **LeetCode** with **10+** contest participations; strong **data structures and algorithms** proficiency
- Solved **520+** problems on **GeeksforGeeks** with a **1700+** coding score; consistent **competitive programming** performance

VOLUNTEER EXPERIENCE

- Co-Head & Technical Content Writer – SAMARTH, TMSL** Kolkata, India
2024 – Present
 - Leadership & Execution:** Led the **IGNITE division**, driving **program management** and **community growth**. Organized **Eduathon** and **Safalya**, ensuring end-to-end execution and strong **stakeholder coordination**.
 - Content & Impact:** Created **technical content** and contributed to **content strategy**, **documentation**, and **outreach**, improving **user engagement** using **data-driven insights**.